Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Real-Time Scalable Plant Disease Classification Using EfficientNet

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Introduction

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1. Plant diseases cause **20–40% annual crop loss** globally.
2. Small-scale farmers lack timely disease diagnosis.
3. Delayed detection leads to rapid disease spread and crop damage.
4. Existing deep learning models are trained on controlled datasets.
5. Real-world conditions differ due to lighting and background variations.
6. Many high-accuracy models are computationally expensive.
7. Such models are unsuitable for affordable mobile devices.
8. This research develops a real-time plant disease classification system.
9. EfficientNet is used to balance:
    1. High accuracy
    2. Computational efficiency
10. Goal: Build a scalable and practical solution for real-world agriculture.

Problem Statement

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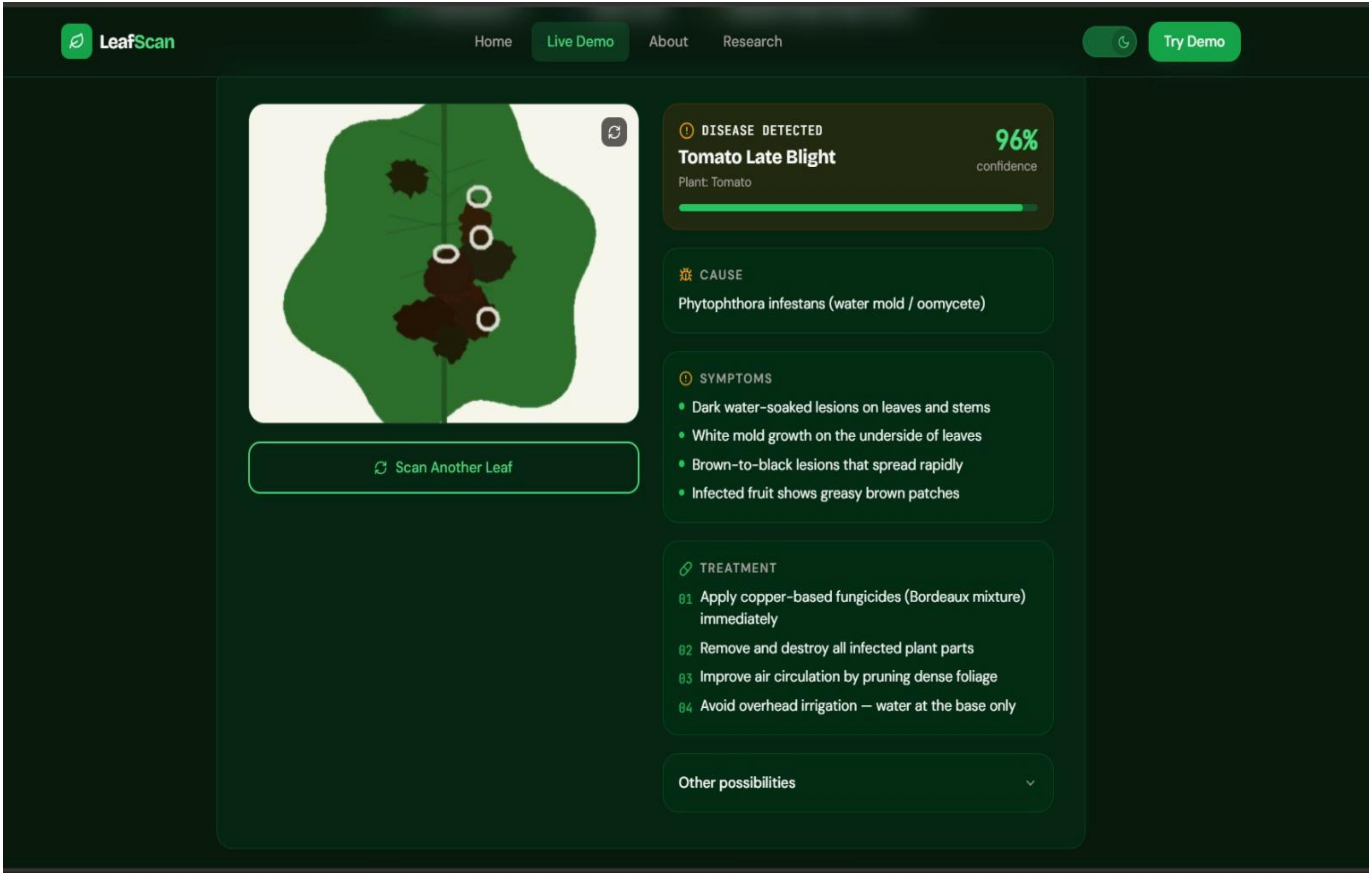
1. Existing deep learning models perform well in laboratory conditions but struggle in real-world environments.
2. Models trained on datasets like PlantVillage often fail due to lighting variations, background noise, and occlusion.
3. Architectures such as ResNet-50 and VGG-16 are computationally expensive for low-cost devices.
4. Dataset imbalance causes biased predictions because some disease classes have fewer samples.
5. There is a need for a model that is accurate, lightweight, and robust for real-world agricultural use.

Objective

1. Develop a high-performance plant disease classification system for real-world agricultural use.
2. Evaluate and compare EfficientNet models (B0–B7) with ResNet-50 and MobileNetV2.
3. Improve model accuracy and generalization using:
    1. Transfer Learning
    2. Data Augmentation
    3. Class-weighted Loss Functions
    4. Cosine Learning Rate Scheduling
4. Enable real-time mobile deployment using TensorFlow Lite with INT8 quantization.
5. Build a complete implementation pipeline with:
    1. FastAPI backend
    2. React frontend
6. Allow users to upload plant images and receive:
    1. Disease predictions
    2. Symptoms information
    3. Treatment recommendations
7. Create a solution that is accurate, practical, and accessible for farmers.

Screenshots



Result

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1. EfficientNet-B4 achieved the best balance of accuracy and efficiency.
2. Achieved **95.9%** test accuracy, outperforming ResNet-50 by **2.1%**.
3. Larger EfficientNet models gave only slight accuracy improvements with much higher complexity.
4. INT8 quantization reduced model size from **76 MB to 22 MB**.
5. TensorFlow Lite model achieved real-time inference (~22 ms).
6. Real-world accuracy was **89.3%** due to domain shift.
7. System provides fast and practical disease prediction for farmers.

Conclusion

This research successfully demonstrates that it is possible to design a plant disease classification system that is both accurate and practical for real-world deployment. EfficientNet-B4 emerged as the most suitable model, offering a strong trade-off between performance and efficiency. The integration of TensorFlow Lite optimization and mobile deployment ensures that the system can operate in real-time on affordable devices, making it accessible to farmers in rural areas.

However, the study also highlights a significant limitation in the form of reduced accuracy on real-world field images, emphasizing the need to address the domain shift problem. Future work should focus on collecting more diverse and representative field data, exploring semi-supervised learning approaches, and incorporating disease localization techniques to improve model reliability and usability. Expanding the dataset to include more staple crops such as wheat, rice, and soybean would further enhance the system's practical impact.

References

1. FAO, *State of Food and Agriculture*, 2021.
2. Savary et al., *Nature Ecology & Evolution*, 2019.
3. Hughes & Salathé, *PlantVillage Dataset*, 2015.
4. Mohanty et al., *Frontiers in Plant Science*, 2016.
5. Tan & Le, *EfficientNet*, ICML, 2019.
6. He et al., *ResNet*, CVPR, 2016.
7. Sandler et al., *MobileNetV2*, CVPR, 2018.

Thank You